

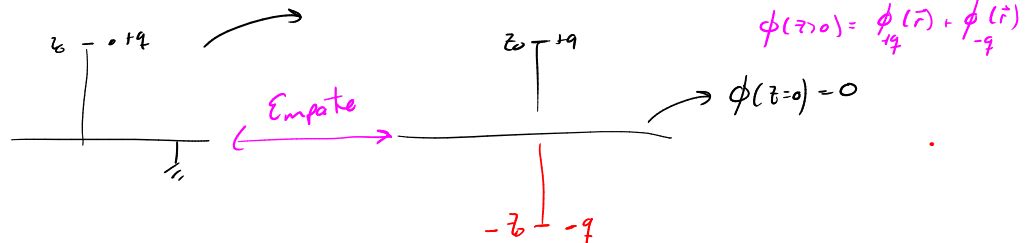
$$\rightarrow \nabla^2 \phi(\vec{r}) = -\frac{\rho(\vec{r})}{\epsilon_0} \quad ; \quad \text{si } \rho(\vec{r})=0 \text{ en } \nabla^2 \phi(\vec{r})=0 \quad \phi|_{\partial V} = \phi_0(\theta)$$

→ Condiciones de frontera $\phi|_{\partial V} = 0 \rightarrow$ Aterrizar

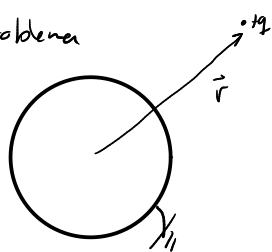
Método de imágenes

• Problema real

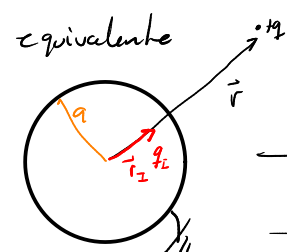
Problema equivalente → Superposición



• Problema



→ Problema



$$\rightarrow r_I = \left(\frac{L}{a}\right) r$$

$$\rightarrow \theta_I = -\frac{r}{a} \theta$$

→ Separación de variables ($\nabla^2 \phi(x,y)=0$)
2D, cartesianas)

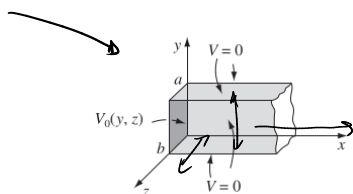
$$\rightarrow \phi(x,y) = \bar{X}(x) \bar{Y}(y) \bar{Z}(z) \rightarrow \frac{\nabla^2 \phi}{\phi} = 0$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\frac{d^2}{dx^2} \bar{X} = +\alpha^2 \bar{X} \rightarrow \exp \rightarrow \bar{X}(x) = A_{\alpha} e^{\alpha x} + B_{\alpha} e^{-\alpha x}$$

$$\frac{d^2}{dy^2} \bar{Y} = -\beta^2 \bar{Y} \rightarrow \text{sen/cos} \rightarrow \bar{Y}(y) = C_{\beta} \text{sen}(\beta y) + D_{\beta} \text{cos}(\beta y)$$

$$\frac{d^2}{dz^2} \bar{Z} = -\gamma^2 \bar{Z} \rightarrow \text{sen/cos} \rightarrow \bar{Z}(z) = E_{\gamma} \text{sen}(\gamma z) + F_{\gamma} \text{cos}(\gamma z)$$



$$\rightarrow \phi(z=0) = \phi(y=0) = \phi(z=b) = \phi(y=a) = 0$$

$$\phi(x=0) = \phi(y,z)$$

$$\phi(x \rightarrow \infty) = 0$$

$$\rightarrow \textcircled{1} \phi(x,y,z) = \bar{X}(x) \bar{Y}(y) \bar{Z}(z) \rightarrow \phi(x,y,z=0) = \bar{X}(x) \bar{Y}(y) (E_{\gamma} \text{sen}(\gamma z) + F_{\gamma} \text{cos}(\gamma z))|_{z=0}$$

$$0 = \bar{X}(x) \bar{Y}(y) F_{\gamma} = 0$$

$$\textcircled{2} \phi(x,y=a,z) = \bar{X}(x) \bar{Z}(z) D_{\beta} = 0$$

$$\textcircled{3} \phi(x,y,z=b) = 0 = \bar{X}(x) \bar{Y}(y) E_{\gamma} \text{sen}(\gamma b) \rightarrow \text{sen}(\gamma b) = 0 \Rightarrow \gamma b = k\pi \rightarrow \gamma = \frac{k\pi}{b}$$

$$\textcircled{4} \phi(x,y=0,z) = 0 = \bar{X}(x) \bar{Z}(z) C_{\beta} \text{sen}(\beta y) = 0 \rightarrow \text{sen}(\beta y) = 0 \rightarrow \beta a = n\pi \rightarrow \beta = \frac{n\pi}{a}$$

$$\rightarrow \phi(x, y, z) = \sum_k \sum_n \left(B_{kn} \right) e^{i\pi x \left(-\sqrt{\left(\frac{k}{b}\right)^2 + \left(\frac{n}{a}\right)^2} \right)} \sin\left(\frac{k\pi}{b} z\right) \sin\left(\frac{n\pi}{a} y\right) \leftarrow$$

$$(6) \quad \phi(x=0, y, z) = \phi_0(y, z)$$

$$\left\{ \sqrt{\frac{z}{b}} \overset{S_k}{\sin\left(\frac{k\pi}{b} z\right)}, \sqrt{\frac{z}{b}} \overset{C_k}{\cos\left(\frac{k\pi}{b} z\right)} \right\}_{k \in \mathbb{Z}}$$

$$(1) \quad \sum_k \sum_n B_{kn} S_k(z) S_n(y) = \phi_0(y, z)$$

$$f: [0, b] \rightarrow \mathbb{R}$$

$$f(z) = \sum_k A_k \left(\sqrt{\frac{z}{b}} \sin\left(\frac{k\pi}{b} z\right) \right) + B_k \cos\left(\frac{k\pi}{b} z\right)$$

$$(2) \quad \sum_k \sum_n B_{kn} S_k(z) S_n(y) S_{k'}(z) S_{n'}(y) = \phi_0(y, z) S_{k'}(z) S_{n'}(y)$$

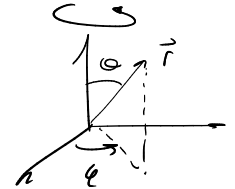
$$\langle S_k, C_k \rangle = \int_0^b S_k(z) C_k(z) dz = 0$$

$$(3) \quad \sum_k \sum_n \int_0^a \int_0^b B_{kn} S_k(z) S_n(y) S_{k'}(z) S_{n'}(y) dy dz = \int_0^a \int_0^b \phi_0(y, z) S_{k'}(z) S_{n'}(y) dy dz$$

$$\langle S_k, S_{k'} \rangle = \delta_{k, k'}$$

$$\sum_k \sum_n B_{kn} \delta_{kk'} \delta_{nn'} = \int_0^a \int_0^b \phi_0(y, z) S_{k'}(z) S_{n'}(y) dy dz = B_{k'n'} \quad k, n = 1, \dots, \infty$$

$$\nabla^2 \phi(\vec{r}) = 0 \quad \text{Es Laplace con simetría azimutal: } \phi(r, \theta)$$



$$\phi(r, \theta) = \sum_{l=1}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta)$$

$$f: [-1, 1] \rightarrow \mathbb{R}$$

$$- P_l(\pm 1) = (\pm 1)^l$$

$$- \text{Paridad de } l \text{ en } \theta$$

$$- \langle P_l, P_{l'} \rangle = \int_{-1}^1 du P_l(u) P_{l'}(u) = \delta_{l, l'} \frac{2}{2l+1}$$

$$\hat{n} \cdot (\vec{E} - \vec{E}^c) \Big|_{\partial V} = \frac{\sigma_{\text{ext}}(\partial V)}{\epsilon_0}$$

$$(\phi^s - \phi^c) \Big|_{\partial V} = 0$$